

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	2023	Date:	

Code of Conduct

I B.Jahnavi / 192325043 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: jahnavi

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. Conversation History: User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage (Give advice and motivate the student). **Constraints:**

1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. Source Sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

Note: "bank" is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.

2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Constraint-Based Coreference Resolution

Text: "John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

a) Referring Expressions and Possible Antecedents

Mention	Type	Candidate Antecedents
John	Proper noun (new entity)	—
Mary	Proper noun (new entity)	—
the park	Definite NP (new entity)	—
He	Pronoun	John, Mary
a ball	Indefinite NP (new entity)	—
She	Pronoun	John, Mary, a ball
it	Pronoun	the park, a ball
The dog	Definite NP (new entity)	—
him	Pronoun	John, the dog
they	Pronoun (plural)	John, Mary, the dog

b) Applying the Resolution Constraints

- Gender/Number agreement: "He"/"him" require a male singular antecedent (eliminates Mary); "She" requires female singular (eliminates John, ball); "it" requires a non-human/inanimate singular antecedent (eliminates John, Mary, dog); "they" requires a plural referent set.
- Recency: among remaining valid candidates, the most recently mentioned entity is preferred — e.g. between "the park" and "a ball" for "it", "a ball" is more recent.
- Semantic compatibility (animacy/action suitability): "play with it" requires a plausible playable object, which excludes "the park" (a location) and favors "a ball"; "chased him" requires an animate target distinct from the subject "the dog" (binding: an object pronoun cannot corefer with the same-clause subject), which excludes the dog and favors John.
- Coherence: the resolved chains must stay consistent with the overall narrative — John and Mary are the two human participants who "go to the park" and later "go home," so "they" naturally continues to refer to the same participant set (optionally extended to include the dog under "all").

c) Constraint Graph (Variables, Domains, Eliminations)

Variable	Domain	Constraint Applied → Elimination	Resolved Value
He	{John, Mary}	Gender(male) eliminates Mary	John
it	{park, ball}	Semantic compatibility eliminates park; Recency favors ball	a ball
She	{John, Mary, ball}	Gender(female) eliminates John, ball	Mary
him	{John, dog}	Binding (object ≠ same-clause subject) eliminates dog	John
they	{John, Mary, dog}	Coherence retains the full recurring participant set	John, Mary, (dog)

d) Final Coreference Chains and Rewritten Paragraph

- Chain 1 (John): {John, He, him}
- Chain 2 (Mary): {Mary, She}
- Chain 3 (the ball): {a ball, it}
- Chain 4 (the dog): {the dog}
- Chain 5 (the group): {they = John, Mary, and the dog}

Rewritten paragraph: "John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home."

e) Priority Order of Constraints and Effect of Relaxing Recency

Priority order: (1) Gender/Number agreement — a hard, categorical filter applied first since it eliminates grammatically impossible candidates outright; (2) Semantic compatibility/animacy — filters out candidates that are grammatically valid but pragmatically implausible; (3) Recency — used as a soft tie-breaker among the candidates that survive the first two filters; (4) Coherence — a global, discourse-level check applied last to ensure the entire set of resolved chains remains consistent across the paragraph.

If the recency constraint were relaxed, the resolver would have no principled way to choose between multiple candidates that pass gender and semantic compatibility filters — for instance, if the semantic-compatibility filter were imperfect, "it" could be left ambiguous between "the park" and "a ball," and the system might default to the first-mentioned or a random candidate, occasionally producing incoherent readings (e.g., implying that Mary wanted to "play with the park"). This shows recency is an important efficient heuristic for disambiguating among otherwise-tied candidates.

Q2. Constraint-Based Dialog Response Generation

User: "I have an important exam tomorrow but I'm not able to concentrate." Required dialog act: Advise + Encourage.

a) Discourse Planning Steps Before Generation

- Content selection: decide which propositions to include — acknowledging the difficulty, giving concrete advice, and offering motivation.
- Discourse relation planning: choose a Cause-Effect structure (e.g., "take a break → concentration improves") to link advice to its expected outcome.
- Entity/reference planning: track and consistently refer to "exam," "concentrate/focus," and "you" across sentences without introducing unrelated entities.
- Lexical choice planning: ensure at least two of the required keywords (focus, break, confident) are naturally incorporated.
- Politeness/tone planning: select an encouraging, positive register and avoid dismissive or negative phrasing.
- Surface realization/length planning: compress the selected content into a fluent 2–3 sentence response.

b) Three Candidate Responses

- R1: "Take a short break to clear your mind — it'll be easier to focus afterward. I'm confident you'll do great on your exam tomorrow!"
- R2: "I know it's hard to concentrate right now, but a quick break can help you refocus before tomorrow's exam. Stay confident — you've prepared well!"
- R3: "Try stepping away for a five-minute break to reset your focus, and remember you have every reason to feel confident about tomorrow's exam."

c) Evaluation and Best Response Selection

Criterion	R1	R2	R3
Entity coherence (exam/focus/you)	Yes	Yes	Yes
Clear Cause-Effect relation	Strong (break → focus)	Moderate	Moderate
Keyword coverage (of 3)	3 (break, focus, confident)	3	2 (break, focus)
Length (2–3 sentences)	2	2	1–2
Politeness/positive tone	Yes	Yes	Yes

R1 is selected as the best response. It satisfies all six constraints, contains a clearly stated Cause-Effect relation ("take a break ... it'll be easier to focus afterward"), includes all three

target keywords, stays within the length limit, and maintains a warm, encouraging tone directly tied to the specific exam mentioned by the user — giving it the strongest overall coherence and constraint satisfaction among the three candidates.

d) Effect of Violating Two Constraints

Consider a response that violates the keyword-inclusion constraint and the logical-consistency constraint simultaneously, e.g.: "Don't worry about the exam, it's not a big deal, just relax." This omits all required motivational keywords, making the advice feel generic and untailored to the dialog act. It also introduces a logical inconsistency: trivializing an exam the user has explicitly called "important" contradicts the user's own framing, which can come across as dismissive rather than supportive. The combined effect significantly degrades the dialog's coherence, appropriateness, and perceived helpfulness, undermining the intended Advise + Encourage dialog act.

Q3. Constraint-Based WSD, Predicate Logic, and Discourse Structure

Source sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

a) Resolving the Ambiguous Word "bank"

The riverbank (landform) sense is selected over the financial-institution sense, justified by the following constraints:

- Contextual co-occurrence: "by the river" strongly co-occurs with the landform sense of "bank."
- Selectional restriction: the predicate "flooded" naturally selects a physical/geographic entity as its argument, which is compatible with a riverbank but not with a financial institution as an abstract organization.
- Absence of financial-domain cues: no co-occurring vocabulary such as "money," "loan," or "account" appears in the sentence to support the financial-institution sense.
- Real-world plausibility: "saved by quick action" (e.g., sandbagging) is a scenario strongly associated with flood defense of a riverbank.

b) Predicate Logic Representation

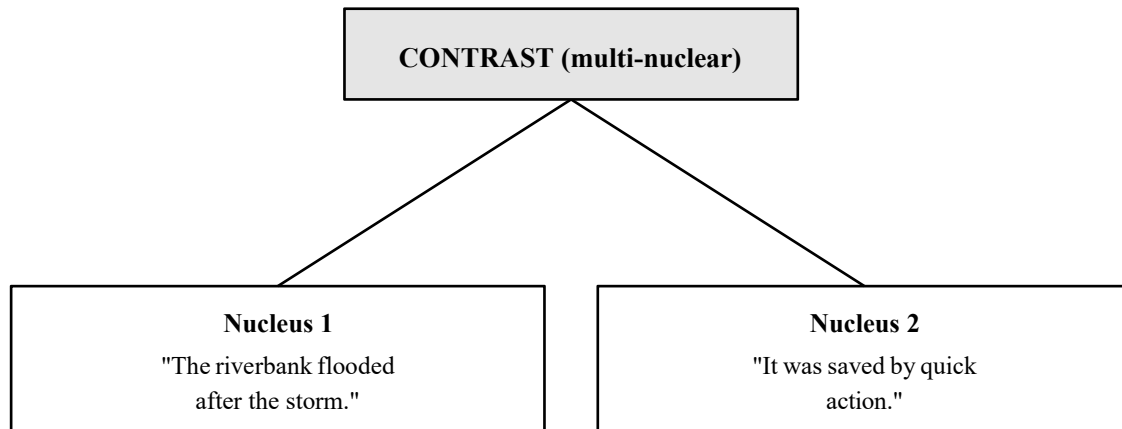
$\exists b \exists r \exists s [\text{Riverbank}(b) \wedge \text{River}(r) \wedge \text{LocatedBy}(b, r) \wedge \text{Storm}(s) \wedge \text{Flooded}(b) \wedge \text{CausedBy}(\text{Flooded}(b), s) \wedge \text{Saved}(b) \wedge \text{Manner}(\text{Saved}(b), \text{QuickAction})]$

Reading: there exists a riverbank b , a river r , and a storm s , such that b is located by r ; b flooded, and this flooding was caused by s ; and b was saved, by means of quick action.

c) Constraint-Satisfying Paraphrase

Simple English paraphrase (disambiguating "bank" by replacing it with an unambiguous term): "The riverbank flooded after the storm, but quick action saved it." This preserves both entities (riverbank, storm), the flood event, the save event, and the Contrast relation between them, with no information loss.

d) RST-Style Discourse Tree (Contrast Relation)



The two clauses are joined by a multi-nuclear Contrast relation: both spans are nuclei of comparable importance, and the rhetorical relation highlights the unexpected turn between the flooding event and its successful mitigation.

e) Advantages of Constraint-Based Approach over Pure Statistical MT

- Explicit disambiguation: constraints such as selectional restrictions and co-occurrence directly resolve polysemous words like "bank," rather than relying purely on training-data frequency, which can favor the statistically dominant (financial) sense even when contextually wrong.
- Guaranteed information preservation: because entities and relations are explicitly tracked (predicate logic, discourse relations), the approach guarantees no silent loss or distortion of meaning — a risk in pure statistical MT, especially for rare senses or low-resource pairs.
- Interpretability and control: each translation decision can be traced back to the specific constraint that produced it, aiding debugging and enabling domain-specific customization (e.g., for legal or technical text).
- Robustness in low-resource settings: constraint-based reasoning does not require large parallel corpora to make correct sense/relation decisions, unlike purely statistical models.
- Trade-off: pure statistical (or neural) MT typically produces more fluent, idiomatic output at scale, so in practice a hybrid of constraint-based disambiguation with statistical/neural generation offers the best balance of accuracy and fluency.